

Artificial Intelligence Assignment 1

Spoken Digit Recognition using GMM and HMM

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1 Introduction

- In this assignment, we worked on building a system that can recognize spoken digits (0–9). During this assignment, we developed a system for recognizing spoken digits (0–9). The goal was to understand and compare two different approaches: Gaussian Mixture Models and Hidden Markov Models, how they can be used for speech recognition problems.
- We trained both models on the data from speakers 1 to 5 and tested the models on data from a different speaker (speaker 6) which was not used during training in order to evaluate how the models generalize to new data. This shows how the models would behave under real life conditions

2 Methodology

2.1 Gaussian Mixture Model (GMM)

- The GMM model just learns the general sound qualities for each digit. It does not attempt to model the order of sounds, but rather just considers the audio as a bag of feature vectors.
- We extracted MFCC features from the audio and modeled each digit with an 8-component Gaussian mixture, and learned the parameters using EM algorithm.
- In very simple terms, GMM learns what each digit “generally sounds like”, without caring about the order in which the sound progresses.

2.2 Hidden Markov Model (HMM)

- The HMM model, however, is more appropriate for sequences. Since the speech changes over time naturally, HMM is more appropriate to our problem.
- We used a 7-state left-to-right HMM. Each state represents a part of the spoken digit (beginning, middle, end). Since this is a discrete HMM, we convert the continuous MFCC features into symbols using K-Means clustering (64 clusters).
- We use Baum-Welch algorithm to train the HMM and we use Forward algorithm to evaluate the HMM. Unlike the GMM, HMM learns to spit a digit step by step.

3 Experimental Results

3.1 Accuracy Comparison

The performance of both models is summarized below:

Table 1: Model Accuracy Comparison		
Metric	HMM (Discrete)	GMM
Validation (Speaker 5)	77.6%	97.00%
Test (Speaker 6)	91.60%	80.00%

It is obvious from the results that HMM is much better than GMM, especially on the unseen test speaker.

3.2 Confusion Matrix Analysis

The confusion matrices for both models are shown below:

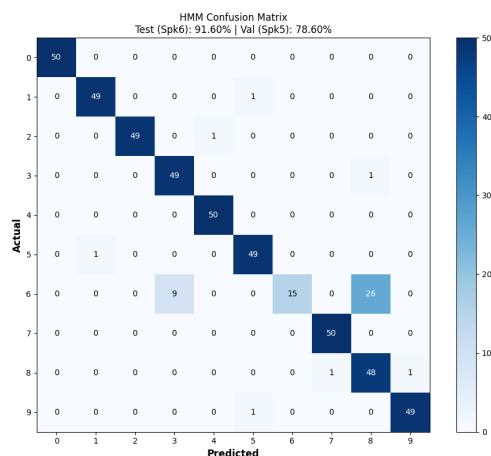


Figure 1: HMM Confusion Matrix

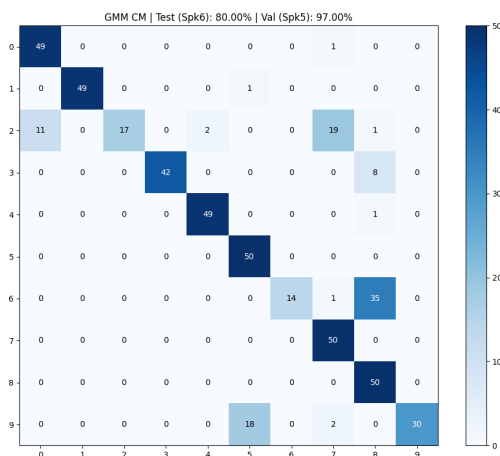


Figure 2: GMM Confusion Matrix

- The HMM confusion matrix has a good diagonal, meaning that most of the predictions are correct. There are still some incorrect predictions.
- So far the most obvious one is the prediction of digit 6 as digit 8. The GMM confusion matrix has more spread out errors. This shows that the GMM has more difficulty with digits that have similar sounds.

3.3 Convergence Analysis

The following graphs show how both models improved during training:

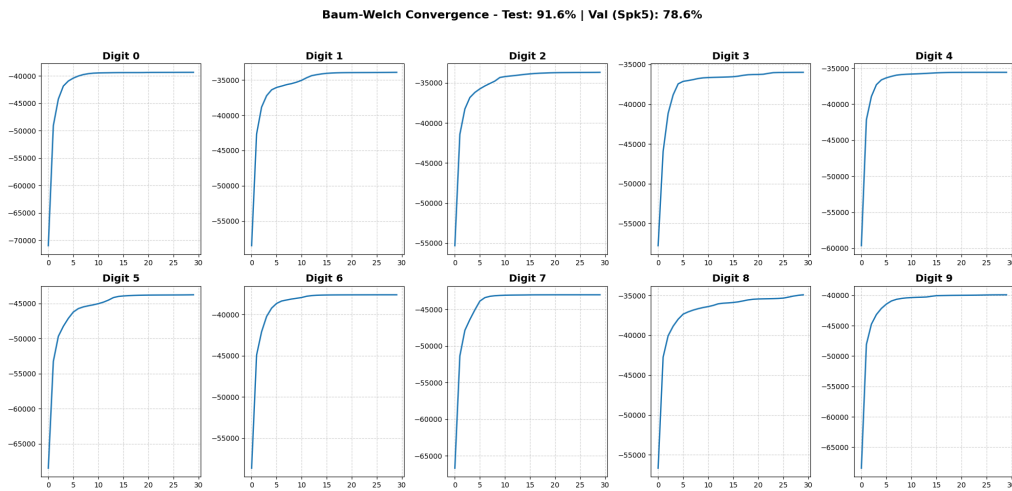


Figure 3: HMM Training Convergence

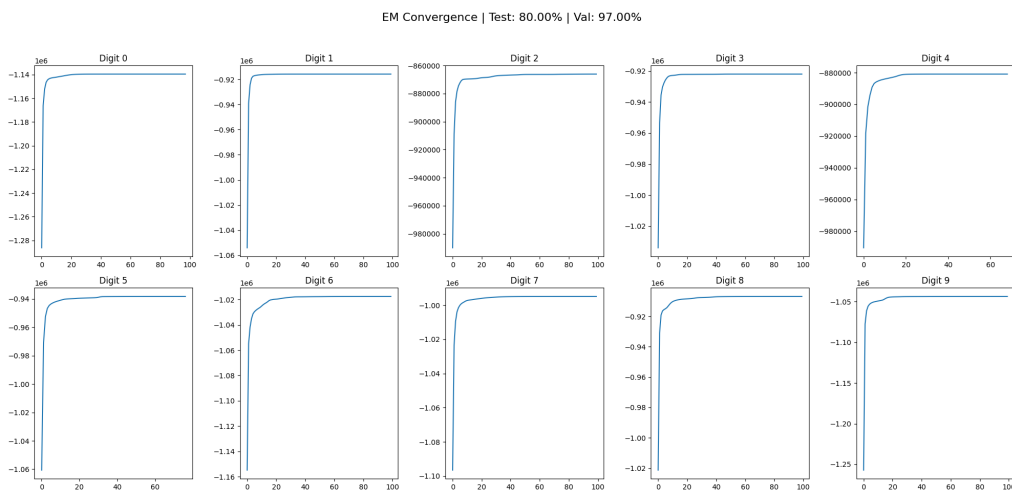


Figure 4: GMM Training Convergence

In both cases, the log-likelihood increases steadily and then levels off after around 20–30 iterations. This is a good sign, as it shows that the models are learning properly and the implementations are stable.

4 Analysis and Insights

4.1 Why HMM Performs Better

Because it takes into account the order of sounds.

- The GMM model doesn't model timing, only cares about their overall characteristics. This means it can get digits that sound close together mixed up.
- Whereas, The HMM model models how sound changes over time. It models the transition between one part of the word to the next. This allows it to discriminate between digits with similar sounds.

4.2 Digit 6 vs Digit 8 Confusion

- An interesting thing to note is the confusion between digit 6 and digit 8 in the HMM results.
- Phonetically speaking, they both end with sharp sounds, so they are not too different.
- Also, since we only used 64 clusters for discretization, some smaller differences may not be able to be captured. Thus, we could have similar representations for both digits.

4.3 Generalization to New Speaker

- The performance degrades significantly when GMM is tested on a different speaker (Speaker 6). This shows that it is quite sensitive to the voice of the training speakers. The HMM performs much better in this situation.
- As it captures more of the structure of speech rather than the voice itself, it generalises well to other speakers.

4.4 Training Behaviour

- Both models converge smoothly during training. The gradual increase in log-likelihood is evidence that the algorithms are correctly implemented and numerically stable.

5 Conclusion

- In this assignment we compared GMM and HMM for spoken digit recognition. The results indicate that HMM is much better than GMM.
- This is because HMM not only model sound features but also the temporal order of these features. GMM can still be used for spoken digit recognition such as speaker identification, but not what is spoken.
- **In general, HMM is better for speech recognition.**